

Anchor Points for Monocular Depth Estimation Under Severe Rolling Shutter Ego-motion

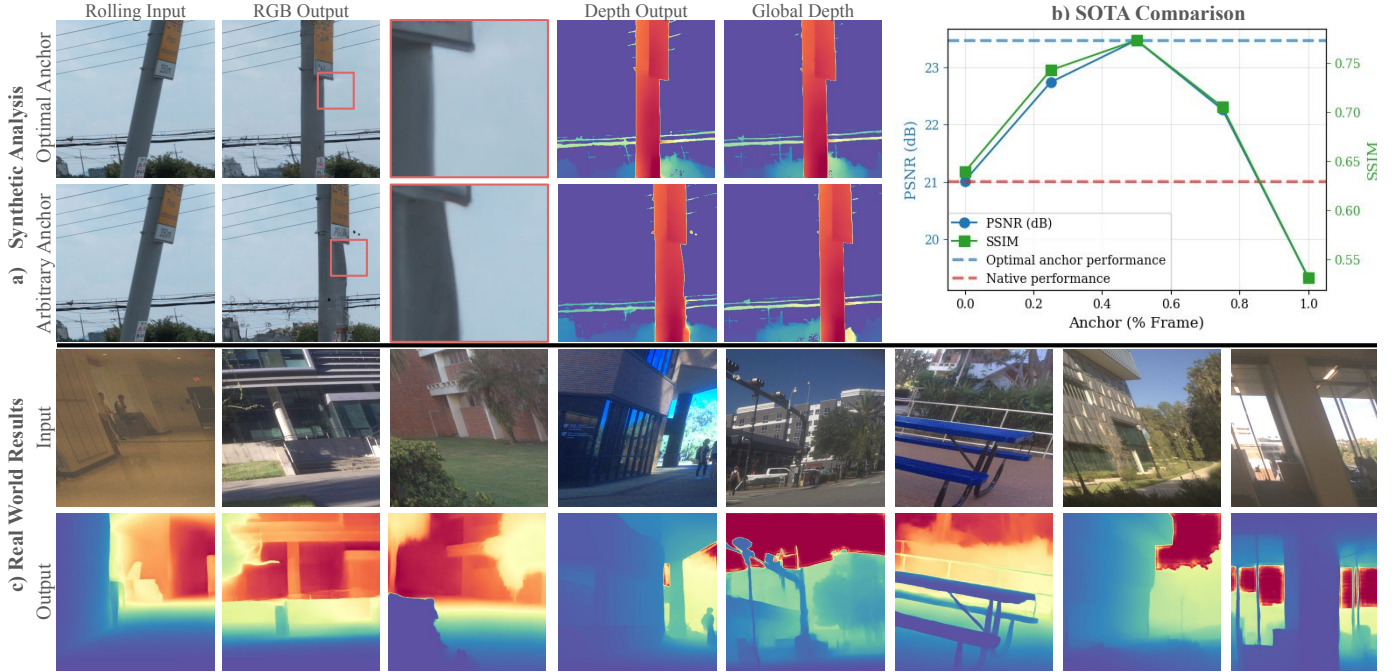


Fig. 1: **Anchor point formulation for rolling shutter correction:** **a)** We introduce the concept of an *anchor point*, which defines the temporal reference used to align rolling shutter observations during correction. **b)** Quantitative analysis on SOTA rolling shutter correction method [1] shows that performance varies systematically with anchor-point selection, providing a significant metric boost around the proposed optimal anchor when compared to native implementation. This modification is **one line of code**, requiring no retraining of the model. **c)** We introduce Visual Foundation Models to rolling shutter correction, providing geometrically consistent monocular depth from large rolling shutter distortions.

Abstract—Rolling shutter (RS) cameras are widely deployed in robotic systems due to their low cost and power efficiency. However, robot ego-motion induces significant RS distortions that degrade geometric consistency and severely challenge visual perception. In this work, we address monocular depth estimation under large rolling shutter distortions caused by ego-motion. We introduce the concept of an anchor point, which establishes a consistent temporal and geometric reference between rolling shutter and global shutter observations. This formulation enables principled adaptation and fine-tuning of deep foundation models for depth estimation under RS distortions. To systematically evaluate the problem, we present a new rolling shutter dataset featuring large motion-induced geometric distortions representative of mobile robotic tasks. Our analysis shows that anchor point selection is a primary factor governing depth accuracy under RS effects. We further characterize anchor configurations for RS-GS data capture and identify stable regimes that yield accurate depth estimates. Using a representative two-stage depth estimation pipeline, we conduct extensive ablation studies demonstrating that anchor-based modeling significantly improves performance, outperforming recent RS correction baselines under strong ego-motion. Finally, we validate our approach on a legged robot platform, demonstrating near real-time monocular depth estimation in the presence of substantial rolling shutter distortions.

I. INTRODUCTION

Rolling shutter (RS) effects arise from row-by-row image readout, causing each scanline to be captured at a different time rather than simultaneously as in global shutter (GS) imaging. In dynamic scenes, this temporal offset induces complex geometric distortions that depend on both camera ego-motion and scene dynamics. RS cameras remain ubiquitous in modern robotic systems (legged platforms, aerial drones, autonomous ground vehicles, mobile manipulation platforms) because they are compact, power-efficient, and cost-effective compared to global shutter alternatives despite their distortions. Since depth estimation is a crucial robot perception task, these platforms must estimate depth even when rapid movements and vibrations amplify RS distortion effects.

Prior work on rolling shutter correction includes geometric methods that estimate camera motion from image cues and learning-based approaches that predict corrected images from RS-GS pairs. Rolling-shutter-aware pose estimation has also been explored in robotics to mitigate downstream effects. Despite these advances, existing methods struggle with single-

frame images under large distortions, often assuming mild motion, requiring extra sensors, or relying on temporal context. Moreover, learning-based methods fix a temporal reference implicitly, yet the impact of this choice on geometric consistency and depth prediction remains underexplored. This lack of exploration causes reliable monocular depth estimation from severely distorted rolling shutter to remain challenging.

A central component of our approach is the **anchor point**, a temporal reference that specifies which global-shutter (GS) frame a rolling-shutter (RS) image is aligned to. While conceptually simple, many modern RSC methods fix this reference implicitly and do not optimize its selection [1], [2]. We show that anchor-point choice is a first-order factor governing correction quality and downstream monocular depth estimation, particularly under severe RS distortions, and that adjusting this reference alone can yield substantial performance gains without retraining (see Fig. 1). For comprehensive evaluation, we introduce a new co-located RS–GS dataset captured with a beam-splitter rig that provides explicit control over anchor points. Using this dataset and a two-stage visual foundation pipeline for RS correction and depth estimation, we demonstrate improved robustness to large distortions and aggressive ego-motion across synthetic and real-world benchmarks.

Overall, our contributions are as follows:

- 1) We introduce the concept of “*anchor point*” as a temporal reference for rolling shutter to global shutter (RS-GS) alignment, and provide a systematic analysis of its role in rolling shutter correction and monocular depth estimation under severe geometric distortions (Fig. 3).
- 2) We present a new co-located RS-GS dataset with explicit anchor-point control, designed to capture realistic robotic motion with large ego-motion effects, and practical imaging artifacts (Fig. 3-5).
- 3) We conduct extensive ablation studies across both synthetic and real-world data to quantify the influence of anchor point selection on reconstruction accuracy and depth consistency (Fig. 6-7, Tables 2-3).
- 4) We develop a two-stage foundation-model-based pipeline and demonstrate its effectiveness for rolling shutter correction and monocular depth estimation under extreme distortions and aggressive ego-motion (Fig. 8, Table 1).
- 5) We validate the proposed system in near-real-time operation with only 400ms end-to-end latency on a legged robot with a co-located sensor setup, including live 3D reconstruction visualization (Fig. 10).

The proposed framework demonstrates promise for real-time RSC by mobile robots under challenging motion conditions. Its validation on a legged robot, subjected to rapid, irregular ego-motion, highlights its viability in real-world applications.

II. RELATED WORKS

Rolling shutter correction. Rolling shutter (RS) correction methods can be broadly categorized by the assumptions they make about motion and available observations. Early single-frame approaches estimate camera motion from visual cues

such as line geometry or motion blur [3]–[5], while multi-frame methods rely on tracking or interpolation across image sequences to recover parametric motion models [6]–[8]. Recent learning-based methods address RS correction using paired RS-GS data [1], [2]. Diffusion-based approaches have shown promise for single-frame RS correction [1], but typically assume moderate distortions or rely on additional sensing such as IMU measurements. In contrast, our work focuses on large RS distortions without external sensor input, emphasizing the role of temporal reference selection in stabilizing correction and downstream depth estimation.

Rolling shutter in robotics. RS distortions are particularly consequential in robotics, where high-speed or rotational motion amplifies RS artifacts. Classical visual odometry (VO) and SLAM pipelines often assume global shutter imagery, leading to degraded pose and structure estimation under RS effects [9], [10]. Continuous-time trajectory representations and rolling-shutter-aware bundle adjustment have been proposed to mitigate these effects [11], [12], while minimal solvers for RS relative pose enable robust RANSAC-based geometry estimation [13], [14]. In visual-inertial pipelines, models that explicitly account for RS readout improve pose and map accuracy under aggressive motion [15]–[17]. This work complements these approaches by focusing on single-frame correction in large-distortion regimes, enabling downstream monocular depth estimation without inertial measurements.

Rolling shutter datasets. Progress in RS correction is closely tied to the availability of representative datasets. Existing datasets such as BS-RSCD [18], Fastec-RS and Carla-RS [2], and Real-RS [1] primarily target mild-to-moderate distortions or rely on synthetic motion profiles. In this work, we introduce a new co-located RS–GS dataset designed to capture large RS distortions arising from realistic robotic motion, with explicit control over temporal alignment through anchor point selection (see Fig. 4). This enables systematic analysis of RS correction and depth estimation under extreme distortions.

Rolling shutter correction and related imaging tasks. RS correction is related to motion deblurring and computational photography techniques that exploit coded exposure or specialized sensing [19]–[21]. Joint RS correction and motion deblurring has been explored [22], [23]. In this work, we assume sharp images without motion blur, consistent with prior RS correction studies [1], [2], [4], and focus on correcting geometric distortions. Unlike approaches that exploit RS artifacts for sensing or coding [24]–[27], our goal is to recover conventional geometry for downstream perception tasks.

Monocular depth estimation under distortions. Monocular depth estimation has seen significant advances through large-scale training and foundation models, including MiDaS [28], DPT [29], AdaBins [30], Depth-Anything [31], UDepth [32], and diffusion methods such as Marigold [33]. While these models generalize well, they assume GS imagery and degrade due to RS distortions. Our work addresses monocular depth estimation by first correcting severe RS artifacts and analyzing how temporal alignment influences geometric consistency.

III. ANCHOR POINT FOR ROLLING SHUTTER CORRECTION

Rolling-shutter (RS) cameras expose rows at different times, producing geometric distortions that depend on motion and row index. Any RSC algorithm must therefore choose a reference time specifying which global shutter (GS) frame the algorithm is attempting to reconstruct. Despite its importance, prior work often selects this arbitrarily (e.g., the first row or midpoint). Below we provide a theoretical definition and analysis of this reference, which we call the *anchor point*. Our analysis explains the importance of anchor point selection and derives why the midpoint $\frac{N}{2}$ is a strong default choice under mild motion assumptions.

Theoretical Summary Our analysis and experiments show the following about anchor-point selection:

- Claim 1: The optimal anchor point lies in a discrete set corresponding to the image rows.
- Claim 2: For any rolling-shutter image, there exists a scene-dependent optimal anchor point that minimizes the total expected distortion.
- Claim 3: Under mild symmetry assumptions on scene motion (e.g., zero-mean deviations about constant velocity), the distortion-minimizing anchor coincides with the temporal midpoint of the exposure, $t_{\text{anch}} = \frac{N}{2}$.

A. Our discrete rolling shutter model

Following [34], the observed RS image can be modeled as an integration over exposure time T of the scene radiance E weighted by the shutter function S :

$$I(x, y) = \int_0^T E(x, y, t) S(x, y, t) dt. \quad (1)$$

We reinterpret each RS image as a discrete sampling from a hypothetical GS video recorded at the same rate as the row readout, so that row y of RS image is sampled from frame y :

$$I(x, y) = G(x, y, y). \quad (2)$$

Thus, correcting RS distortion amounts to choosing a GS frame anchor point t_{anch} and warping all rolling shutter rows toward that frame.

B. Theoretical claims

Claim 1. *The optimal anchor point lies in a discrete set*

Proof Sketch of Claim 1. Each row is captured at integer time index y , so choosing

$$t_{\text{anch}} = k \in 1, \dots, N \quad (3)$$

forces row k to be perfectly aligned with its GS counterpart ($D_k(k) = 0$, where D can be any non-negative distortion metric). Non-integers do not provide this same property of perfect alignment. Under standard temporal smoothness, the expected distortion of other rows depends only on $|y - k|$, yielding

$$\mathbb{E}[D_y(t)] = h(|y - t|), \quad h(0) = 0, \quad h(u) > 0 \quad \forall u > 0. \quad (4)$$

Thus the set of viable anchors is the set of row indices. $\square$

Corollary 1.1. *Because the feasible set contains only N elements, the optimal solution can be obtained by a grid search over candidate anchors.*

Claim 2 (General Optimal Anchor). *For any rolling-shutter image, there exists an optimal global-shutter anchor point t_{anch}^* that minimizes the total expected distortion.*

Proof Sketch of Claim 2. Let X_n denote the horizontal position of a tracked structure in row n , and $X(t)$ be the GS reference position at anchor t . By defining a general distortion metric $D(t) = \sum_{n=1}^N |X_n - X(t)|$, we can write

$$X_n = vn + \alpha_n, \quad X(t) = vt, \quad (5)$$

where v is nominal linear motion and α_n encodes accelerations, biases, or asymmetries. Then

$$D(t) = \sum_{n=1}^N |(vn - vt) + \alpha_n|. \quad (6)$$

Minimizing $D(t)$ over $t \in \{1, \dots, N\}$ defines the general optimal anchor optimization pipeline as:

$$t_{\text{anch}}^* = \arg \min_{t \in \{1, \dots, N\}} \sum_{n=1}^N |(vn - vt) + \alpha_n|. \quad (7)$$

$\square$

Claim 3. *Under mild symmetry assumptions on scene motion (e.g., zero-mean deviations about constant velocity), the distortion-minimizing anchor coincides with the temporal midpoint of the exposure, $t_{\text{anch}} = \frac{N}{2}$*

Proof Sketch of Claim 3. For the derivation above, if the deviations satisfy $\mathbb{E}[\alpha_n] = 0$ for all n , then

$$\mathbb{E}[D(t)] = \sum_{n=1}^N |vn - vt|, \quad (8)$$

which is minimized by the median of $\{1, \dots, N\}$. For evenly spaced rows, this is the temporal midpoint $t_{\text{anch}}^* = \frac{N}{2}$. $\square$

Corollary 3.1. *The midpoint $\frac{N}{2}$ arises as a special-case solution under zero-mean, symmetric motion statistics.*

These claims show that anchor-point selection is a fundamental factor in RS correction. Our method serves as a strong instantiation demonstrating these effects, while the dataset provides a controlled benchmark for systematic evaluation.

C. Anchor points as a method-agnostic improvement

It is important to note that the **anchor-point selection is not specific to our pipeline**, but is implicitly present in all RS correction methods that map an RS image to a single GS reference. To demonstrate this, we revisited RS-Diffusion [1], which predicts a dense flow field that maps each RS row to a GS image. In its original formulation, the predicted flow is implicitly anchored to the first row.

Without retraining or modifying the network, we re-centered the predicted flow so that the displacement is zero at the image midpoint rather than at the top row, effectively changing the



Fig. 2: Highlights of the Real Dataset. Each pair of rows shows the rolling shutter capture and the corresponding global shutter capture. The dataset will be released to the public.

anchor point from $t_{\text{anch}} = 0$ to $t_{\text{anch}} = \frac{N}{2}$. This simple post-processing step substantially improves reconstruction quality, seen in Fig. 1 over three metrics.

This result provides strong empirical evidence that anchor-point choice is a first-order factor in rolling shutter correction. Even state-of-the-art pretrained models benefit significantly from correct temporal alignment, highlighting anchor point selection as a paramount decision in dataset creation.

IV. METHODOLOGY

A. Dataset Creation and Analysis

To evaluate rolling shutter correction under large distortions, we introduce three datasets containing paired RS and GS images. These datasets emphasize motion profiles and distortions commonly encountered in robotic systems, and provide explicit anchor-point control for systematic study.

SYN1: Synthetic Benchmark. *SYN1* contains RS–GS pairs rendered in Blender [35]. Each single-view scene (table fan, ceiling fan, propeller plane) includes two GS images for different anchor points and eight RS variants with increasing distortion, producing $(8 + 2) \times 200 = 2000$ images per scene. Randomization in lighting, colors, and textures ensures appearance diversity. To prevent overfitting, we hold out specific blade orientations (15° intervals) for testing.

SYN2: Realistic RS Simulation. *SYN2* introduces realistic motions and textures while maintaining full control over RS effects. RS frames are synthesized from the high-speed X4K1000FPS dataset [36] by row-wise stitching of consecutive GS frames. This method allows the selection of multiple anchor points for a single RS image. Each RS image corresponds to five GS frames, defining five anchor points per sequence. The dataset includes **22,040** GS frames and **4,408** RS images, covering a variety of motions and scene types.

REAL: Collocated RS–GS Capture. *REAL* is a real-world dataset captured using a **dual-camera collocated system** (Fig. 3), where rolling-shutter and global-shutter cameras are beam-splitter-aligned and hardware-synchronized. Varying trigger delays provides explicit control of the anchor point, producing multiple RS–GS pairs per scene. The system is

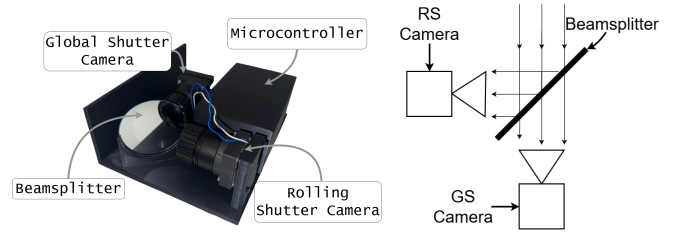


Fig. 3: Collocated RS–GS camera setup used for REAL. Hardware synchronization allows explicit control of anchor points for each capture. This device will be provided as a demo if accepted.

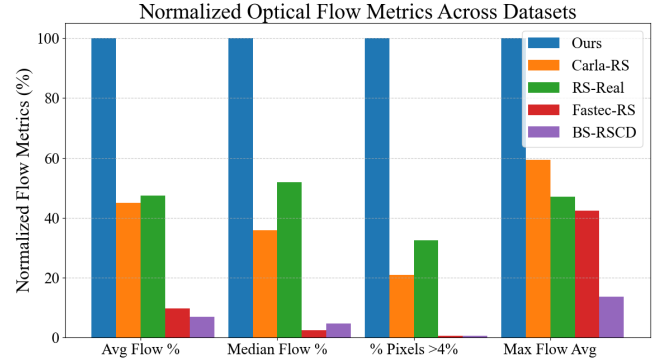


Fig. 4: Optical-flow-based motion statistics for RS datasets. **REAL** exhibits the largest motion, making it suitable for evaluating large-distortion RS correction. The percentage of pixels greater than 2%, 6%, and 10% for our dataset was also over double all other datasets compared. **No other dataset matches our egocentric motion, allowing us to model the effects of legged robots and other mobile systems.**

attachable to a Puppy-Pi robot for legged-robot experiments. The dataset contains 40k image pairs at 300×300 resolution (20k outdoor, 10k indoor) with additional images for anchor-point ablations. Optical-flow analysis (Fig. 4) shows that **REAL** exhibits the largest motion among RS datasets, making it suitable for evaluating large-distortion RS correction and depth estimation.

B. Integration of Vision Foundation Models

To address RS correction and monocular depth estimation, we leverage pre-existing vision foundation models (VFMs). Our pipeline consists of two sequential stages:

- 1) **RS Correction Stage:** We adopt a Marigold U-Net [37] trained to map RS images to corresponding GS frames. The model operates on RGB images and focuses on removing row-wise distortions while preserving structure.
- 2) **Monocular Depth Estimation Stage:** Corrected RS frames from the first stage are passed to a transformer-based Depth-Anything-V2 [31]. This stage produces dense depth maps, benefiting from the reduction in RS-induced geometric distortions. The depth predictions are

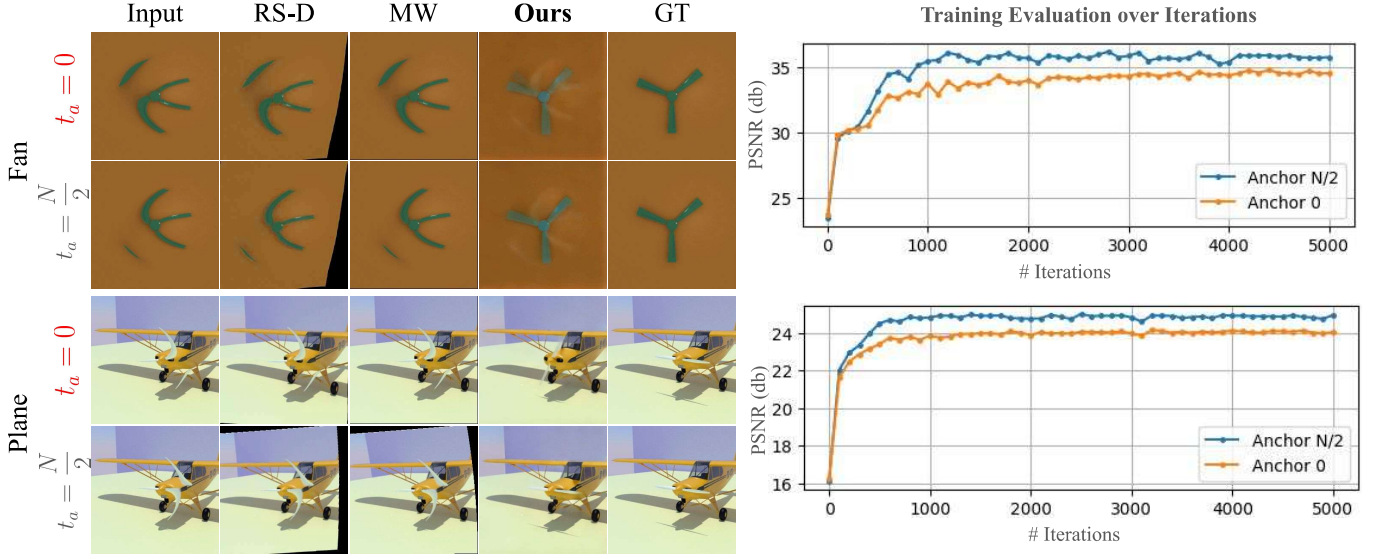


Fig. 5: Visualization of the models run on synthetic data (SYNTH1). Each column corresponds to a method, and each row to a scene (Fan, Plane). Each scene shows the input with two anchors ($t_a = 0$ and $t_a = \frac{N}{2}$), compared against RS-Diffusion [1], Manhattan World [4], our model outputs, and ground truth. The PSNR curves on the right show training performance over iterations for each scene.

compared with the GS depth maps produced by a frozen Depth-Anything-V2 teacher model for evaluation.

Pipeline Details. The stages are trained sequentially: first, the Marigold U-Net is trained for RS correction; second, the corrected outputs are used to fine-tune the depth model. We experimented with different VFM configurations and found that sequentially applying two distinct VFMs yields better depth accuracy than either model alone.

Training Setup. Both stages are trained using our RS-GS paired datasets (SYN1, SYN2, REAL). Standard image augmentation is applied, and losses include MSE loss for RS correction and built-in scale-invariant loss for Depth-Anything-V2. All other architectural components (e.g., U-Net layers, transformer blocks) remain unchanged from the original VFMs, highlighting that improvements derive from task-specific training and anchor-point alignment.

Purpose of VFM Integration. We use Vision Foundation Models (VFMs) to leverage strong priors for RGB reconstruction and monocular depth estimation, allowing us to isolate and analyze rolling shutter distortions and anchor-point effects. Depth-Anything is employed as a second-stage evaluator to assess structural and geometric consistency of the corrected outputs, rather than focusing on high-frequency photometric artifacts. This enables evaluation of rolling shutter behavior at a geometric level relevant to depth estimation.

V. EXPERIMENTAL ANALYSES

Our experiments were designed to validate the theoretical motivation for anchor points, evaluate VFM’s on RSC, and analyze how different design choices affect rolling shutter correction and depth estimation quality.

A. Anchor Points for Controlled Synthetic Scenes

We conduct single-scene experiments on the synthetic dataset SYN1 to isolate the effect of anchor-point selection. Scenes exhibit rotation about the image center, suggesting an optimal anchor of $t_a = \frac{N}{2}$.

Six diffusion models were trained across three scenes, comparing optimal ($t_a = \frac{N}{2}$) and suboptimal ($t_a = 0$) anchor points. Only models trained with the optimal anchor successfully corrected all three scenes, confirming our theoretical predictions (Fig. 5, Tables I).

We further compare against **Manhattan World** [4] and **RS-Diffusion** [1], which struggle under extreme distortions, while our VFM pipeline preserves structure and continuity.

TABLE I: Comparisons of different anchor points with the first stage of our two-stage architecture on single-scene datasets (Ceiling Fan, Table Fan, Plane). The models were evaluated using PSNR and SSIM on intermediate RGB images. $t_a = \frac{N}{2}$ achieves the highest performance across all scenes.

Anchor	Ceiling Fan		Table Fan		Plane	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
$t_a = 0$	34.93	0.971	26.82	0.831	24.21	0.808
$t_a = \frac{N}{2}$	36.21	0.986	27.57	0.842	25.02	0.841

B. Anchor Points in General Realistic Scenes

Next, we evaluated the full two-stage pipeline, RS correction followed by monocular depth estimation.

Synthetic Results. On the X4K-based dataset [36], we trained models using different anchor-point choices spanning the exposure window. Performance consistently peaked near the temporal midpoint, with $t_a = \frac{N}{2}$ yielding the highest PSNR across training (Fig. 7). Visual comparisons show cleaner

TABLE II: Comparisons of different anchor points with the first stage of our two-stage architecture on multi-scene datasets (X4K and Collocated). These comparisons were done on the intermediate RGB images, and $t_a = \frac{N}{2}$ consistently achieves the best or comparable performance.

Anchor	X4K		Collocated	
	PSNR	SSIM	PSNR	SSIM
$t_a = 0$	15.594	0.578	20.238	0.639
$t_a = \frac{N}{4}$	17.291	0.630	20.292	0.640
$t_a = \frac{N}{2}$	17.733	0.629	21.778	0.674

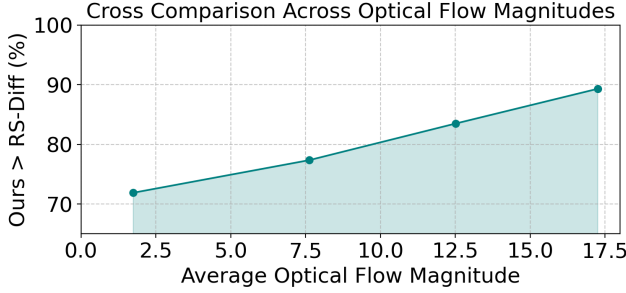


Fig. 6: Relative performance of our model compared to RS-Diffusion across optical flow magnitudes. Each bin represents the percentage of images where our model achieves higher PSNR. The performance gap widens with increasing motion, indicating that our approach is more robust to large rolling shutter distortions.

reconstructions and more consistent depth than prior RS correction methods (Fig. 8).

Real-World Results. On the novel **REAL** dataset, our pipeline consistently outperformed baseline RS correction models and Depth-Anything-V2 in both RGB fidelity and depth accuracy (TABLE. III). PSNR improvements are most pronounced in high-motion sequences, demonstrating robustness to large RS distortions (Fig. 6).

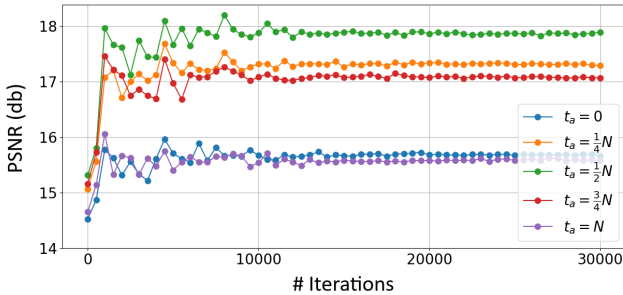


Fig. 7: Comparison of PSNR in the five models described in Sect. 5.B. Each anchor point is represented by a different color on the graph. We validated our theory for anchor points for general scenes. Performing a grid search on the possible anchor point location yielded different networks with varying accuracies. This experiments supported the use of the anchor point $\frac{N}{2}$ for the real-world scenes.

C. Ablation Study: Model Configuration

We evaluated three configurations: **one-stage**, **parallel two-stage**, and **orthogonal two-stage** (Table III). **One-stage** models (Marigold [37], Depth-Anything [31], EcoDepth [39]) jointly learn de-rolling and depth but struggle with structural consistency. **Parallel two-stage** models decouple the tasks sequentially using identical architectures; this improves stability but provides limited gains. **Orthogonal two-stage** models combine distinct architectures (Marigold U-Net for de-rolling and Depth-Anything-V2 for depth) and achieve the highest PSNR, SSIM, AbsRel, and δ_1 . Even untuned orthogonal baselines outperform parallel two-stage designs, indicating that sequential specialization of tasks is a pivotal factor behind performance improvements. Fine-tuning further improves results, confirming that the pipeline effectively leverages VFM for RSC and Monocular Depth Estimation.

D. Ablation Study: Anchor Points in Other RSC Algorithms

We evaluated RS-Diffusion [1] across a sweep of anchor points and found that centering the predicted flow at the temporal midpoint ($t_{\text{anch}} = \frac{N}{2}$) consistently yields the best reconstruction on both synthetic and real RS images. This confirms that mid-exposure alignment improves performance across different RSC methods without retraining, highlighting the generality of the anchor-point insight (Fig. 1).

E. Effectiveness on Legged Robots

Legged robots are subject to rapid and irregular motion, which can induce significant rolling shutter distortions and consequently degrade depth estimation performance. To validate our method in this real-world setting, we collected 1,000 images from the collocated system attached to Puppy-Pi, where the global shutter camera serves as ground truth for evaluation. The model was further modified for faster inference by reducing floating-point precision to 16-bit and decreasing denoising steps from 50 to 10. These changes reduce average runtime on the pipeline from 2850 ms to 349 ms. Highlighted visualizations and metrics from data captured on puppy-pi can be seen in Fig. 9.

VI. CONCLUSION

We studied monocular depth estimation under severe rolling shutter distortions induced by robot ego-motion and identified anchor-point selection as a first-order factor governing geometric consistency and depth accuracy. By explicitly modeling this temporal reference, we showed that existing rolling shutter correction methods and vision foundation models can be significantly improved without architectural changes or re-training. Using a new co-located RS-GS dataset and extensive synthetic and real-world experiments, we demonstrated robust depth estimation under large distortions and aggressive motion, including deployment on a legged robot with a latency under 400ms. These results highlight temporal alignment as a critical design consideration for rolling shutter perception in mobile robotic systems.

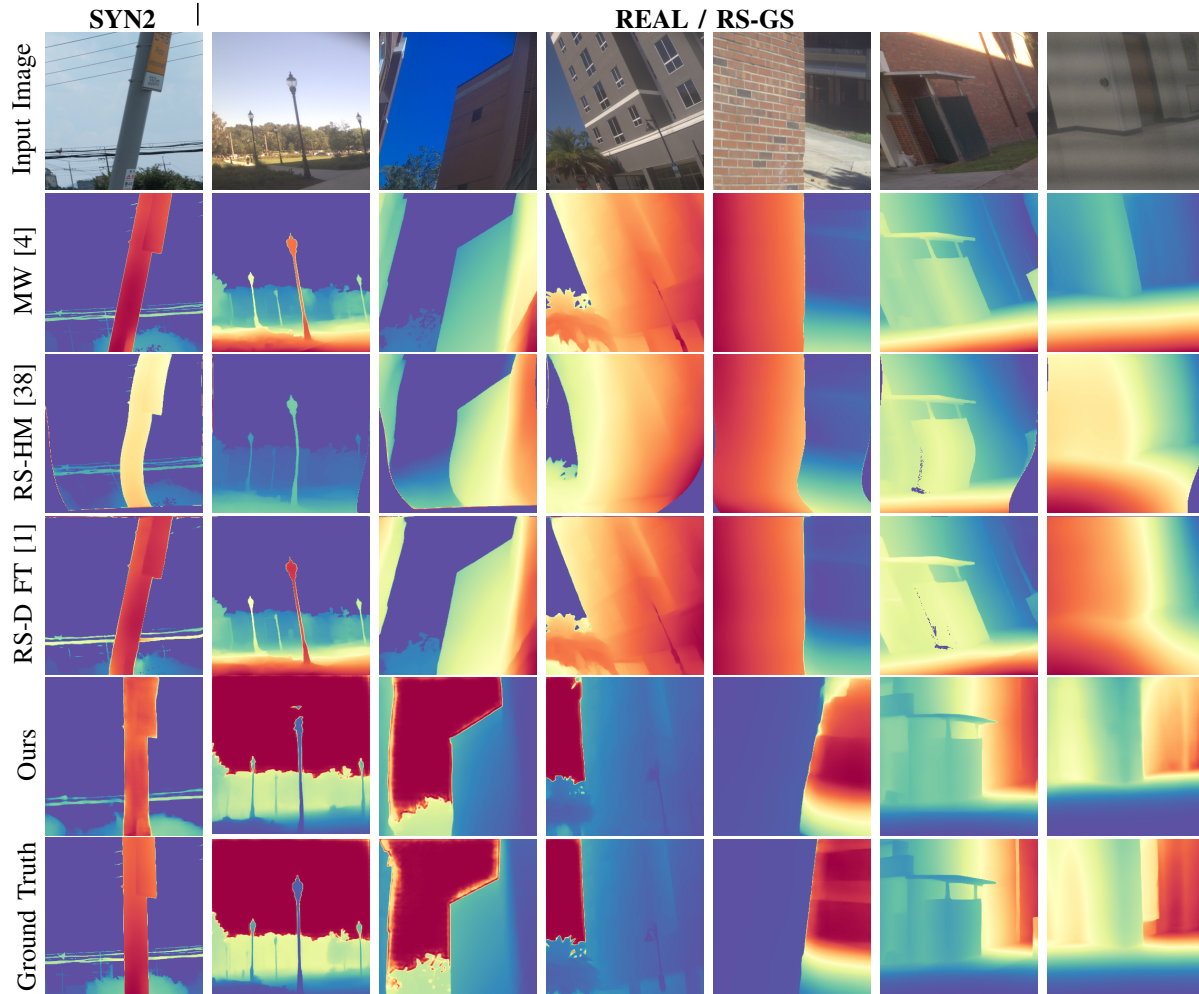


Fig. 8: Qualitative results on real scenes. From top to bottom: input image, two baseline comparison outputs, fine-tuned comparison output, our model’s output, and ground truth. Only our method produces accurate depth, demonstrating the two-stage approach preserves scene structure. **Note Column 5**, where naive affine-based rectification does not work, since the ground truth has a slanted structure. Additional visualizations and comparisons can be seen in the supplementary video.

TABLE III: Comparisons of PSNR, SSIM, AbsRel, and δ_1 on the testing dataset used in Sect. 5.2. Our fine-tuned, two-stage architecture (U-Net, Transformer) performs better on validation than any other combination of the architectures. Values that are not possible to record (intermediate RGB metrics for one-stage architecture) are represented with NA.

Architecture	Method	Intermediate RGB		Depth Map	
		PSNR	SSIM	AbsRel	δ_1
One Stage	Marigold [33]	NA	NA	6.047	0.257
	Depth-Anything [31]	NA	NA	0.655	0.757
	Eco-Depth [39]	NA	NA	0.488	0.676
Comparisons	RS-Diff [1]	19.47	0.433	0.904	0.190
	RS-Diff Fine-tuned [1]	21.52	0.469	0.802	0.214
	Deep RS-HM [38]	20.75	0.456	0.788	0.446
	Manhattan [4]	21.41	0.454	0.621	0.623
Parallel Two Stage	[33]- [33]	22.87	0.494	3.22	0.382
Orthogonal Two Stage	[33]- [39]	22.87	0.494	0.506	0.649
	Baseline [33]- [31]	22.87	0.494	0.377	0.773
	Fine Tuned (ours)	22.87	0.494	0.191	0.842

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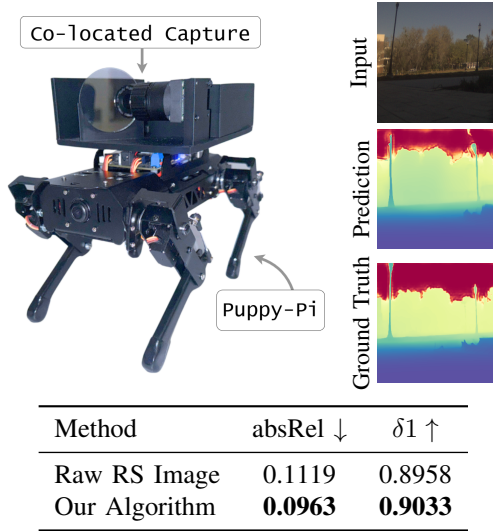


Fig. 9: Left: schematic illustration of the RS effect on the Puppy-Pi. Right: input, prediction, and ground truth for a single frame. Bottom: dataset-averaged depth metrics comparing raw RS input and corrected output.

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